WEEK 12 – AUTOMOBILE ANALYTICS DASHBOARD

VARNIKASRI G

E25DA051

B.SC DS & AI

1. Introduction

The Automobile Analytics Dashboard project was developed using Microsoft Power BI to analyze and visualize automobile-related data. The main purpose of this project is to understand vehicle performance, pricing, engine specifications, body styles, and drive-wheel categories through interactive visualizations.

The cleaned automobile dataset from Week 11 was reused for this project. Various Power BI visuals, including KPI Cards, Scatter Plot, Comparison Matrix, Interactive Tooltips, and Slicers, were created to develop an informative and interactive dashboard.

2. Objectives

- To analyze automobile data using Power BI.
- To display important automobile metrics using KPI Cards.
- To compare horsepower and vehicle prices using a Scatter Plot.
- To analyze vehicle prices based on body style and drive wheels.
- To display vehicle technical specifications using Interactive Tooltips.
- To create an interactive dashboard using slicers and filters.

3. Dataset Description

The Automobile dataset used in this project was prepared and cleaned during Week 11. It contains information about different automobile manufacturers, vehicle specifications, engine characteristics, fuel types, and prices.

Important columns used in this project include:

- Make
- Price
- Horsepower
- Engine Size
- Highway MPG
- Curb Weight
- Body Style
- Drive Wheels
- Fuel Type

The cleaned dataset was loaded into Power BI for visualization and analysis.

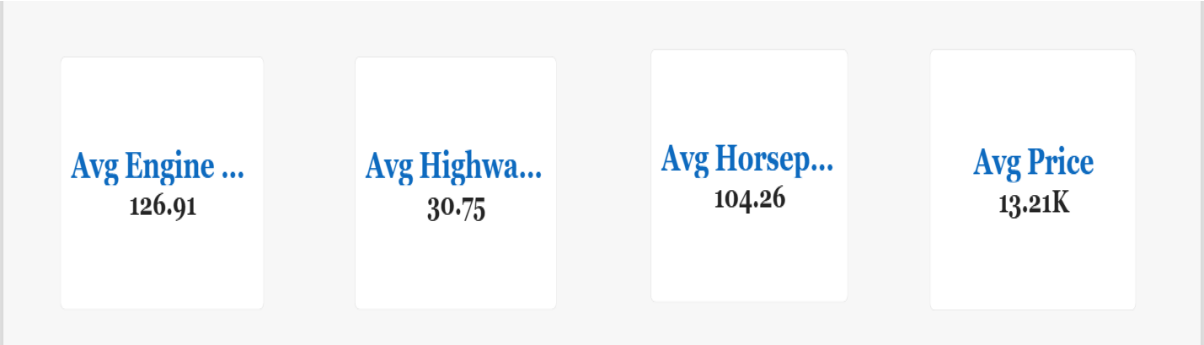
4. Creating KPI Cards

Four KPI Cards were created to display important automobile statistics.

The following measures were used:

1. Average Engine Size – 126.91
2. Average Highway MPG – 30.75
3. Average Horsepower – 104.26
4. Average Price – 13.21K

These KPI Cards provide a quick overview of the automobile dataset and help users understand the average values of important vehicle attributes.



5. Scatter Plot – Horsepower vs Price

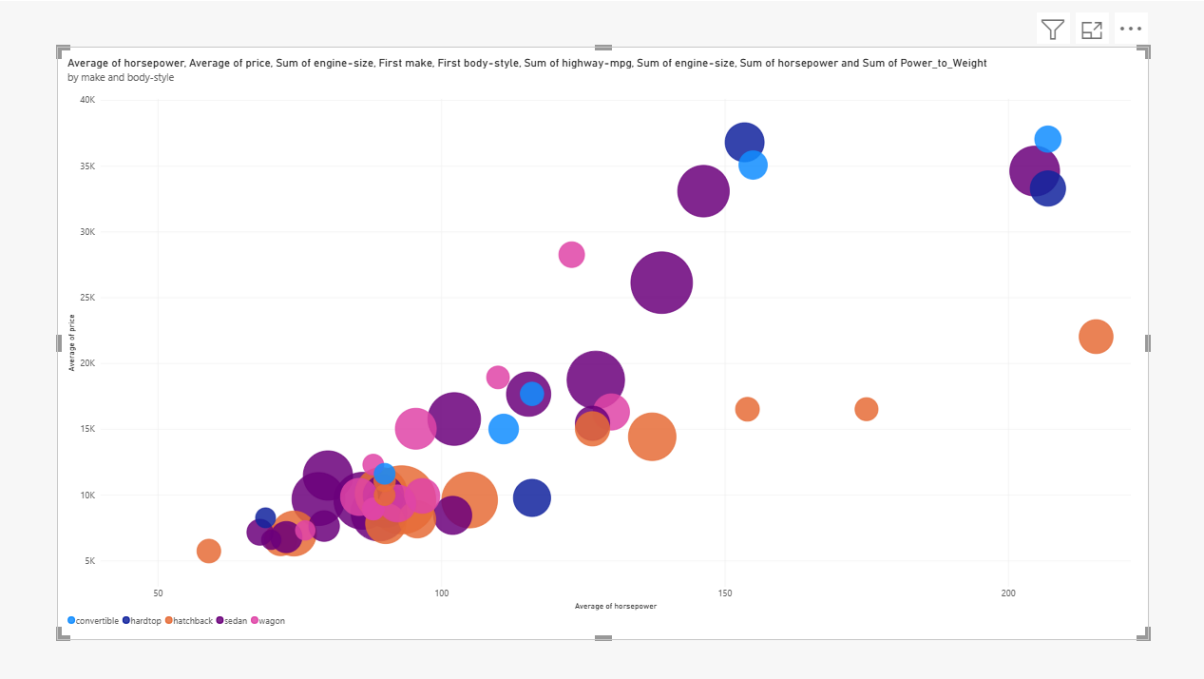
A Scatter Plot was created to analyze the relationship between vehicle horsepower and price.

The following fields were assigned:

- X-Axis: Horsepower
- Y-Axis: Price
- Size: Engine Size
- Legend: Body Style
-

Average aggregation was applied to the horsepower and price fields.

This visualization helps compare automobile prices with horsepower and engine size. Different colors represent different body styles, making the comparison easier.



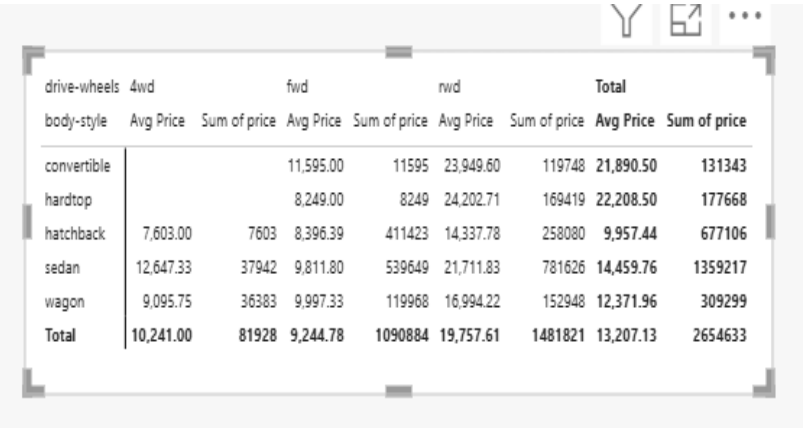
6. Comparison Matrix – Body Style vs Drive Wheels

A Matrix visual was created to compare automobile prices based on body style and drive-wheel categories.

The following fields were used:

- Rows: Body Style
- Columns: Drive Wheels
- Values: Price

The matrix helps users compare different vehicle categories and understand how automobile prices vary across body styles and drive-wheel types.



drive-wheels	4wd		fwd		rwd		Total	
body-style	Avg Price	Sum of price	Avg Price	Sum of price	Avg Price	Sum of price	Avg Price	Sum of price
convertible			11,595.00	11595	23,949.60	119748	**21,890.50**	**131343**
hardtop			8,249.00	8249	24,202.71	169419	**22,208.50**	**177668**
hatchback	7,603.00	7603	8,396.39	411423	14,337.78	258080	**9,957.44**	**677106**
sedan	12,647.33	37942	9,811.80	539649	21,711.83	781626	**14,459.76**	**1359217**
wagon	9,095.75	36383	9,997.33	119968	16,994.22	152948	**12,371.96**	**309299**
Total	**10,241.00**	**81928**	**9,244.78**	**1090884**	**19,757.61**	**1481821**	13,207.13	2654633

7. Interactive Tooltips – Vehicle Technical Specifications

A separate report page named **Vehicle Tooltip** was created to display detailed automobile information.

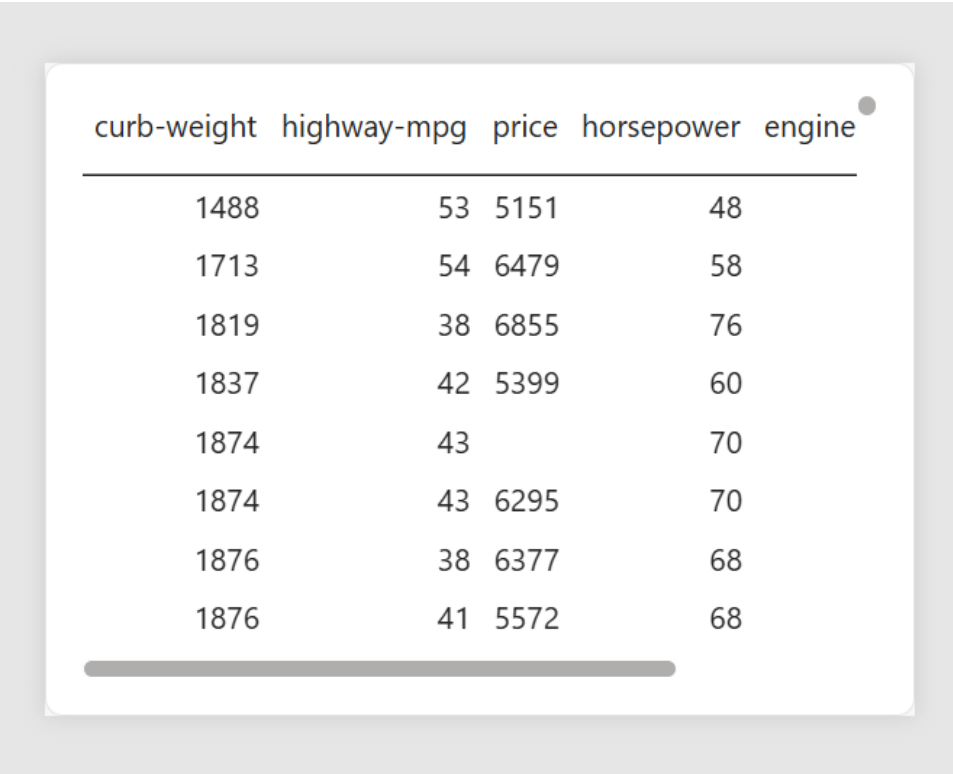
The tooltip page contains the following fields:

- Make
- Price
- Horsepower
- Engine Size
- Highway MPG
- Curb Weight

A table visual was used to display the technical specifications. The tooltip page was configured as a report page tooltip and connected to the main Scatter Plot.

When the user moves the mouse pointer over a vehicle point in the Scatter Plot, the corresponding vehicle details are displayed automatically.

This feature provides additional information without displaying too many details on the main dashboard.



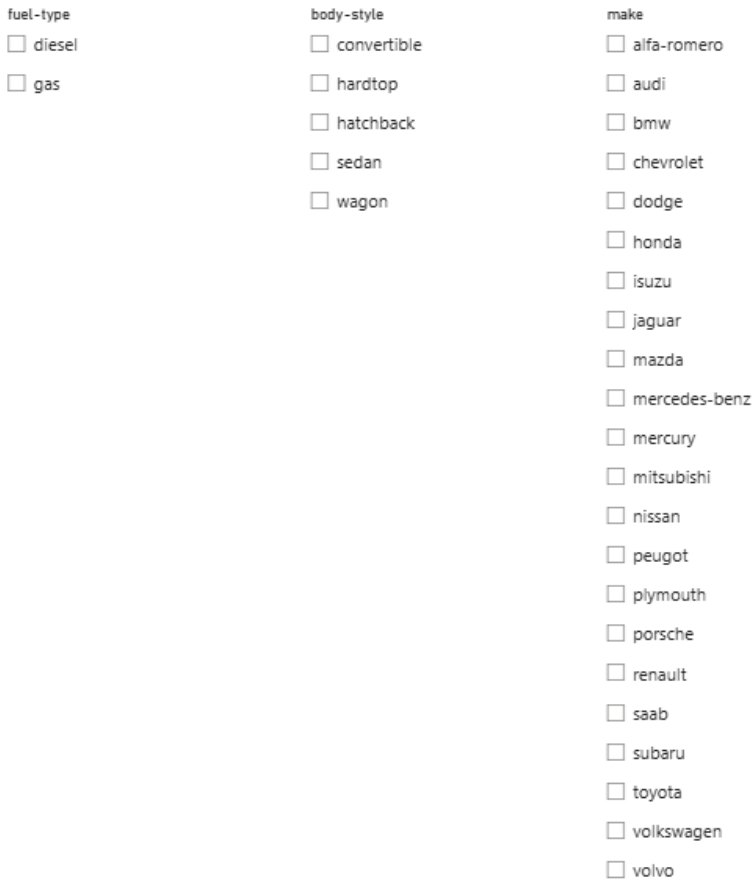
curb-weight	highway-mpg	price	horsepower	engine
1488	53	5151	48	
1713	54	6479	58	
1819	38	6855	76	
1837	42	5399	60	
1874	43		70	
1874	43	6295	70	
1876	38	6377	68	
1876	41	5572	68	

8. Interactive Slicers

Three slicers were added to make the dashboard interactive.

- Make – Filters automobiles based on manufacturer.
- Fuel Type – Filters automobiles based on fuel category.
- Body Style – Filters automobiles based on vehicle body type.

These slicers allow users to explore the automobile data according to their selected categories. The dashboard visuals update based on the selected filters.



9. Final Dashboard Design

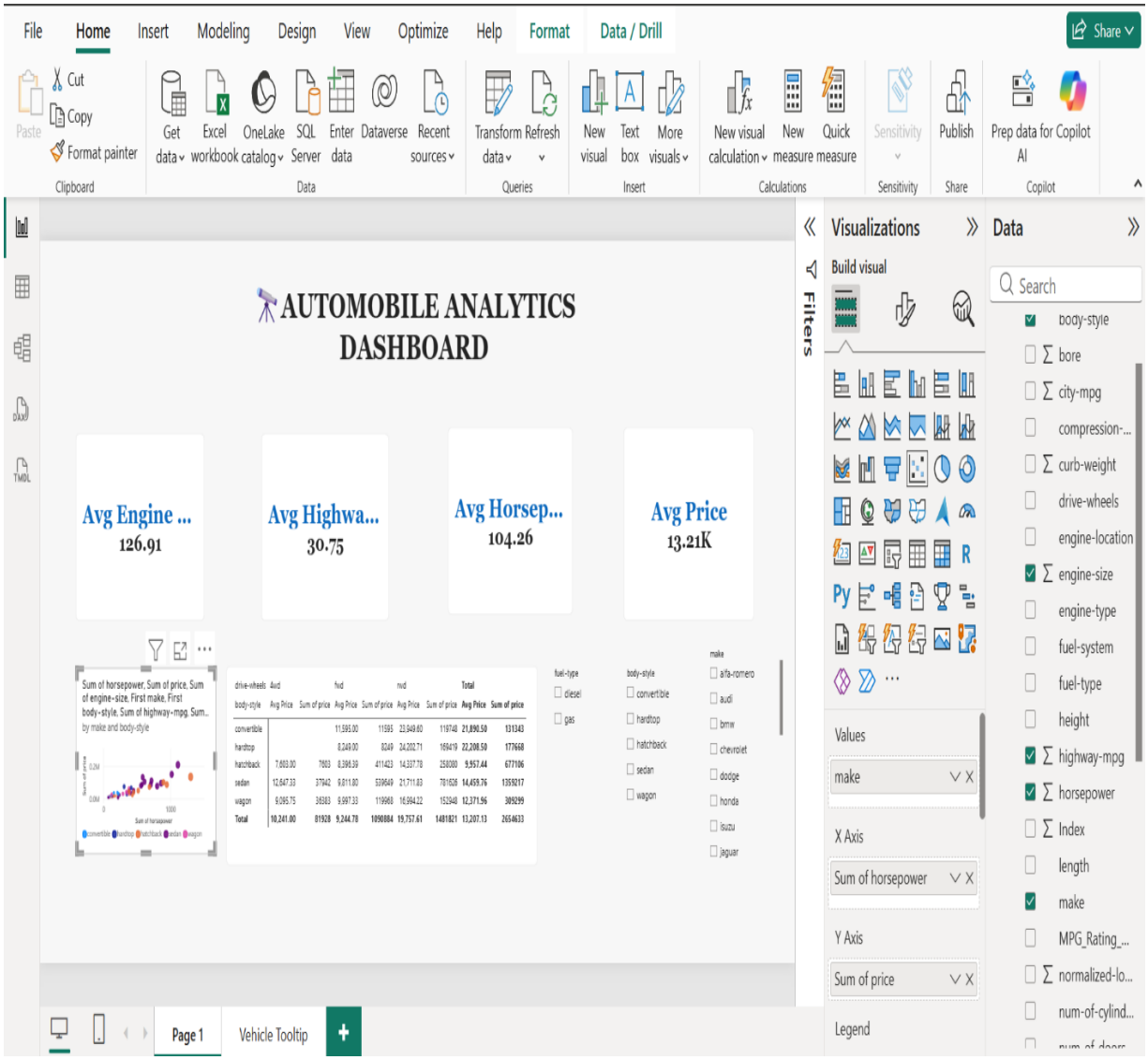
All the created visualizations were arranged together on a single report page named **Automobile Dashboard**.

The final dashboard includes:

- Dashboard Title
- Four KPI Cards
- Scatter Plot

- Comparison Matrix
- Make Slicer
- Fuel Type Slicer
- Body Style Slicer

The dashboard was designed with a clear layout to make the information easy to understand. The separate Vehicle Tooltip page was also connected to the Scatter Plot to provide additional technical details.



10. Project Outcome

The Automobile Analytics Dashboard was successfully developed using Microsoft Power BI. It provides an interactive platform to analyze automobile pricing, horsepower, engine size, fuel types, body styles, and drive-wheel categories.

KPI Cards provide a quick summary of important metrics, while the Scatter Plot and Comparison Matrix help users compare automobile characteristics. Interactive Tooltips provide detailed vehicle specifications, and slicers allow users to filter the dashboard dynamically.

11. Conclusion

This project provided practical experience in creating interactive dashboards using Microsoft Power BI. It helped in understanding data visualization, KPI creation, comparison analysis, report page tooltips, and interactive filtering.

The Automobile Analytics Dashboard presents automobile information in a simple and organized manner. Overall, this project improved my knowledge of Power BI and strengthened my data analysis and visualization skills.